4.3 Sequences of Transformations

Lesson Introduction

Benchmarks	MA.912.GR.2.2 Identify transformations that do or do not preserve distance. MA.912.GR.2.3 Specify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure.	Learning Targets	• I can identify transformations that preserve distance. • I can specify a sequence of transformations that will map a given figure onto another congruent figure.

Benchmark Coherence	Building On	Working Toward
	MA.8.GR.2.1 Given a preimage and image generated by a single transformation, identify the transformation that describes the relationship.	N/A

Essential Questions	• How can you determine if a transformation preserves distance? • How can you use a sequence of transformations to map a given figure onto another congruent figure?
Lesson Narrative	In the lesson, students will apply what they have learned about translations, rotations, and reflections in the prior two lessons to begin piecing them together in a sequence of transformations.

Component Summary				
	4.3.1 Warm-Up (~5 min)	Fill-in-the-blank statements *(SE p.* 190)	4.3.4 Your Turn! (10 min)	Independent practice on describing transformations and writing transformations using algebraic descriptions *(SE p.* 194)
	4.3.2 Guided Instruction (15-20 min)	Guided instruction on how to sequence multiple transformations *(SE p.* 191)		
	4.3.3 Guided Practice (10 min)	Completing a table with the transformation and algebraic description for a specified mapping *(SE p.* 194)	4.3.5 Wrap-Up (~5 min)	Self-reflection on the level of understanding of each learning target *(SE p.* 195)

Resources	Glossary	Materials	Additional Preparation
	Key Terms: N/A Additional Terms: • Origin • x-axis • y-axis	• Patty paper	N/A

	Common Misconceptions	Things to Consider
Teacher Tips	• Students may apply the wrong sign when completing a reflection. • Students may write the wrong algebraic description for a different rotation type than the one intended. • Students may struggle to determine the correct sequence to complete the mapping.	• Consider elements of instruction from Unit 1 resources that can activate prior knowledge during this lesson. Both Unit 1 and the activities in this lesson purposefully position students as the sense makers in order to make meaningful connections between their work with figures in Unit 1 and transformations in this lesson.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Compare and Connect During 4.3.2 Guided Instruction activity, students receive guided instruction on the sequence of transformations. While guiding students through the activity, it will be important to share teacher thinking so that students can make the connection that the activity intends.	**MA.K12.MTR.5.1: Patterns and Structure** Throughout the core activities of this lesson, students will be applying their prior knowledge of isolated transformations to specifying a sequence of transformations to map a given figure onto another congruent figure.
Differentiate Instruction	Students who struggle with individual transformations will likely carry that over to sequences of transformations. In the beginning, consider breaking the sequences apart into separate transformations before combining into one complete problem. Providing this entry point will highlight student understanding of individual transformations, which can be used to celebrate what has been learned and to determine remediation steps for what still needs work.	

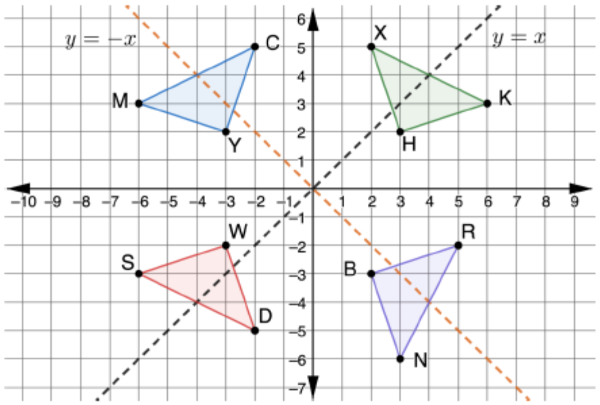
Lesson Notes:

4.3.1 Warm-Up: Bell Work

Component Overview: *Students will complete a fill-in-the-blank activity that assesses their prior learning with reflections.*

4.3.1 Warm-Up: Bell Work

Triangles MCY, KXH, NRB, and SWD are shown on the coordinate grid.



Complete each statement.

1. When ΔMCY is reflected across *the $y-axis$*, it will map onto ΔKXH.
2. When ΔNRB is reflected across $y=x$, it will map onto ΔMCY.
3. When ΔSWD is reflected across the x-axis, it will map onto ΔMYC.

This activity will provide insight into what information students have retained from the prior lessons. Watch how students perform on this Warm-Up to get an idea for which students may need more support during this lesson.

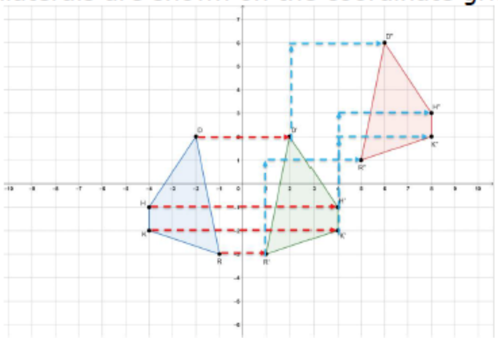
How does the line of reflection impact what happens to the image? What are some ways that this happens?

4.3.2 Guided Instruction: Sequence of Transformations

Component Overview: *Students will be introduced to a sequence of transformations. This activity is designed to connect their prior learning of isolated transformations to sequences of transformations.*

4.3.2 Guided Instruction: Sequence of Transformations

1. Quadrilateral $RKHD$ was transformed to create quadrilateral $R'K'H'D'$. Then, quadrilateral $R'K'H'D'$ was transformed to create quadrilateral $R''K''H''D''$. The quadrilaterals are shown on the coordinate grid.



   a. On the graph, show the mappings of the vertices of the first transformation.
   Shown on graph in red.

   b. On the graph, show the mappings of the vertices of the second transformation.
   Shown on graph in blue.

   c. Complete the table.

	Quadrilateral $RKHD$ Transformed to Quadrilateral $R'K'H'D'$	Quadrilateral $R'K'H'D'$ Transformed to Quadrilateral $R''K''H''D''$
Written Description	*Reflection across the y-axis.*	*Vertical translation up 4 units. Horizontal translation right 4 units.*
Algebraic Description	$(x, y) \rightarrow (-x, y)$	$(x, y) \rightarrow (x + 4, y + 4)$

If available, consider using physical manipulatives or patty paper to connect each step of the sequence for students to connect individual transformation to the overall sequence.

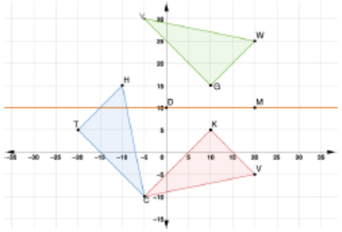
Pay close attention to how students write their algebraic description. Many students will be able to choose the correct reflection but may inaccurately write in the algebraic description.
Ask students to notice the direction of the vertices when naming the preimage and image for a reflection.
For example, $RKHD$ is named clockwise, starting at R, while $R'K'H'D'$ is named counterclockwise, starting at R'. With a reflection, the direction of the vertices changes to the opposite direction. Relate this to the change in direction when taking a picture with a front-facing selfie camera versus the back camera.

Students first encountered transformations that preserve distance in Unit 1. The figures were on the plane, not in the coordinate plane. The act of mapping the vertices could be extended to mapping all points, and because all of the points can be mapped to corresponding points, the figures are congruent.

4.3.2 Guided Instruction: Sequence of Transformations (continued)

d. Explain how to know if a sequence of transformations preserves distance.
Answers will vary. Sample response: Each of the vertices on the quadrilaterals are the same distance apart after each transformation.

2. Triangles HTC, VKC, and WGY are shown on the coordinate grid. Triangle HTC is the pre-image. Triangle VKC is the result of the first transformation, and ΔWGY is the result of the second transformation.



a. Describe the first transformation.
ΔHTC rotated $90°$clockwise about point C to create ΔVKC.

b. Use patty paper to show that the transformation preserved distance.

c. Describe the second transformation.
ΔVKC reflected over $\overleftrightarrow{DM}$ $(y = 10)$ to create ΔWGY.

d. Use patty paper to show that the second transformation preserved distance.

Position students to make connections between their exploration of congruence in Unit 1 and their use of patty paper for mapping figures on the coordinate plane. Consider utilizing activities, videos, or visuals from Unit 1 as additional scaffolding for students to make those connections.

For question 2b, allow the students to try it on their own or with a partner. Based on how familiar your students are with patty paper, consider giving them general instructions and example of using the patty paper to make a rotation.

4.3.2 Guided Instruction: Sequence of Transformations (continued)

e. Work with your partner to complete the statement.

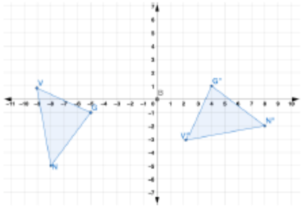
Because the [○ center of rotation / ○ line of reflection] of [○ ΔHTC / ○ ΔVKC] is [○ $(0,0)$, / ○ $(-5,-10)$, / ○ $y=0$, / ○ $y=10$,] an algebraic description [○ can / ○ cannot] be used.

f. When Judeline and Carmen completed the statement above, they had different correct answers. Write the second correct answer.
Option 1: center of rotation, ΔHTC, $(-5,-10)$, cannot
Option 2: line of reflection, ΔVKC, $y=10$, cannot

To use an algebraic description of a rotation the center of rotation must be $(0,0)$ *(the origin)*.

To use an algebraic description of a reflection the line of reflection must be x-axis, y-axis, $y=x$, or $y=-x$.

3. Triangles NVG and $N''V''G''$ are shown on the coordinate grid. Triangle NVG was transformed to create $\Delta N''V''G''$.



a. Explain which features of the two triangles can be used to determine that the transformations were rigid motions.
Answers will vary. Sample response: The lengths of the sides of the triangle remain the same.

There are two correct answers for question 2e. The answer chosen in the first box determines the correct choice for the second and third boxes. Observe which response students choose and validate all correct answers through discussion of questions 2e and 2f. The answer to question 2f will depend on which options were chosen in question 2e. Both correct answers are shown under question 2f on the key.

- *Why can't an algebraic description be used if the center of rotation $(-5,-10)$ is a point on the triangle?*
- *Is this a rule that would apply to other triangles as well? Why or why not?*

Consider the use of an anchor chart for the blue boxes in question 2f. A T-chart could be used to illustrate algebraic description usage based on the type of transformation, including an example.

Discussion questions to consider:

- *Why would the lengths of the sides of the triangles remain the same? How do you know?*
- *How can you determine if a rotation has occurred?*
- *How can you tell if the rotation was clockwise or counterclockwise?*

4.3.2 Guided Instruction: Sequence of Tranformations (continued)

b. Adele told Kiriam that she knows one of the rigid motions is a rotation. Discuss with your partner which clues in the location of the vertices Adele used to draw her conclusion. Write a summary of your discussion.
Answers will vary. Sample response: the location of the vertices are rotated from the pre-image.

c. Work with your partner to determine a sequence of transformations that includes a rotation about the point B so that when applied, ΔNVG will map onto $\Delta N''V''G''$. Complete the table.

Transformation	Mapping	Algebraic Description
Sample Answer 1 Step 1: Vertical translation down 3 units and horizontal translation right 6 units. *Sample Answer 2 Step 1: Rotation of 90° counterclockwise about point B.*	$\Delta NVG \rightarrow \Delta N'V'G'$	*Sample Answer 1 Step 1:* $(x, y) \rightarrow (x+6, y-3)$ *Sample Answer 2 Step 1:* $(x, y) \rightarrow (-y, x)$
Sample Answer 1 Step 2: Rotation of 90° counterclockwise about point B. *Sample Answer 2 Step 2: Vertical translation up 6 units and horizontal translation right 3 units.*	$\Delta N'V'G' \rightarrow \Delta N''V''G''$	*Sample Answer 1 Step 2:* $(x, y) \rightarrow (-y, x)$ *Sample Answer 2 Step 2:* $(x, y) \rightarrow (x+3, y+6)$

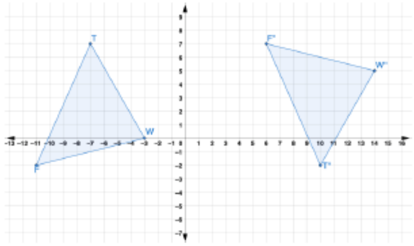
Consider modeling the thinking for one of the transformations and then having students determine the algebraic description. Repeat this again, but have students determine the transformation and then model the algebraic description. For the second mapping, release students to try it with a partner before reviewing the answers.
Ask students to notice the direction of the vertices when naming the preimage and image for a rotation. For example, NVG is named clockwise, starting at N, and $N'V'G'$ is also named clockwise, starting at N'. With a rotation, the direction of the vertices does not change direction.

4.3.3 Guided Practice: Sequence of Transformations

Component Overview: *Students work in a teacher-supported setting to complete a table with the transformation and algebraic description for a specified mapping scenario. For this activity, release students to try this question on their own or with a partner and then come back together to review as a whole group.*

4.3.3 Guided Practice: Sequence of Transformations

1. Triangles TWF and $T''W''F''$ are shown on the coordinate plane.



Determine a sequence of transformations that will map ΔTWF onto $\Delta T''W''F''$. Complete the table.

Transformation	Mapping	Algebraic Description
Reflection across the x-axis.	$\Delta TWF \rightarrow \Delta T'W'F'$	$(x, y) \rightarrow (x, -y)$
Vertical translation up 5 units and horizontal translation right 17 units.	$\Delta T'W'F' \rightarrow \Delta T''W''F''$	$(x, y) \rightarrow (x + 17, y + 5)$

Consider using some of the time during this component to work with a small group of students who need additional support based on their responses and performances throughout this lesson.

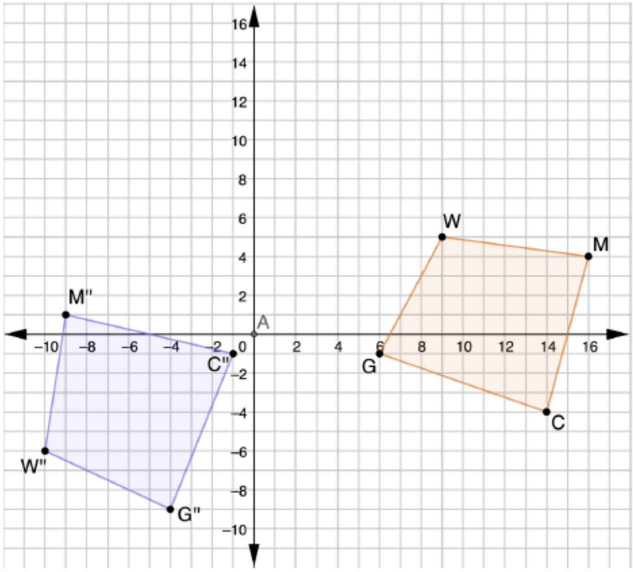
For students who finish early, consider having them write an algebraic description of a transformation and then trade with a partner to write the description of the transformation. Repeat the process in the reverse order.

4.3.4 Your Turn!

Component Overview: *Students will engage in independent practice on describing transformations and writing transformations using algebraic descriptions. For this activity, students should work independently to practice what they've learned about sequences of transformations.*

4.3.4 Your Turn!

1. Quadrilateral $MCGW$ was transformed to create quadrilateral $M''C''G''W''$.



a. Describe the sequence of transformations that will map quadrilateral $MCGW$ onto quadrilateral $M''C''G''W''$.
Answers will vary. Rotation of $90°$ counterclockwise about point A. Vertical translation down 15 units then horizontal translation 5 units left.

b. Write the sequence of transformations using algebraic descriptions.
Answers will vary.
$(x, y) \rightarrow (-y, x)$
$(x, y) \rightarrow (x - 5, y - 15)$

Student performance on this task can provide insight into what students may need additional support with:

- *Question 1a correct, question 1b incorrect:*
  - *Students can describe the correct transformation but may not understand how to accurately write in an algebraic description.*
- *Question 1a incorrect, question 1b correct:*
  - *While this is rare, it would indicate that students can represent the transformation algebraically but cannot describe the transformation.*
- *Question 1a incorrect, question 1b incorrect:*
  - *While students were unable to get the correct transformation, it is important to review whether or not their algebraic description matches their written description.*

4.3.5 Wrap-Up: Reflection

Component Overview: *Students will complete a self-reflection on their level of understanding of each learning target. This activity is designed to be done independently to gather data on students' self-assessments of their level of understanding from this lesson.*

4.3.5 Wrap-Up: Reflection

1. Check the box that describes your understanding of each learning target.

- ☐ I can identify transformations that preserve distance and I am ready for the next lesson.
- ☐ I need some help identifying transformations that preserve distance. One area in which I need help is ______________________________.
- ☐ I need more support in knowing how to identify transformations that preserve distance. I am concerned I will not be ready to move on. One area in which I need help is ______________________________.

- ☐ I can specify a sequence of transformation that will map a given figure onto another congruent figure.
- ☐ I need some help specifying a sequence of transformation that will map a given figure onto another congruent figure. One area in which I need help is ______________________________.
- ☐ I need more support in knowing how to identify transformations that preserve distance. I am concerned I will not be ready to move on. One area in which I need help is ______________________________.

Answers will vary.

Notice the open-ended responses for the second and third options for each question. If students struggle to identify one area, consider having them choose a problem or an activity from the lesson in this blank.